AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 11 · STUDENT LESSON

Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Fit the question. Store the leaves. Trace the branch.

Distinguish an inspectable depth-one learner from inference with a supplied rule.

Keep one habit: Six complete training rows → branch counts → fixed validation comparison → optional pruning fixture → independent cutoff5trace.

MAIN LESSON · SECTION 1

Separate fitting a small tree from following its branches

> A stored threshold can be traced without showing how training selected it.

Learning goals

I can fit a **root** question and **leaf prediction**s from **training** examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Retrieve baseline and held-out reasoning

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.
- Why does a cutoff with zero **training** errors still need validation on separate examples?

End goal

By the end, I can explain decision trees using a traced example and a stated limitation.

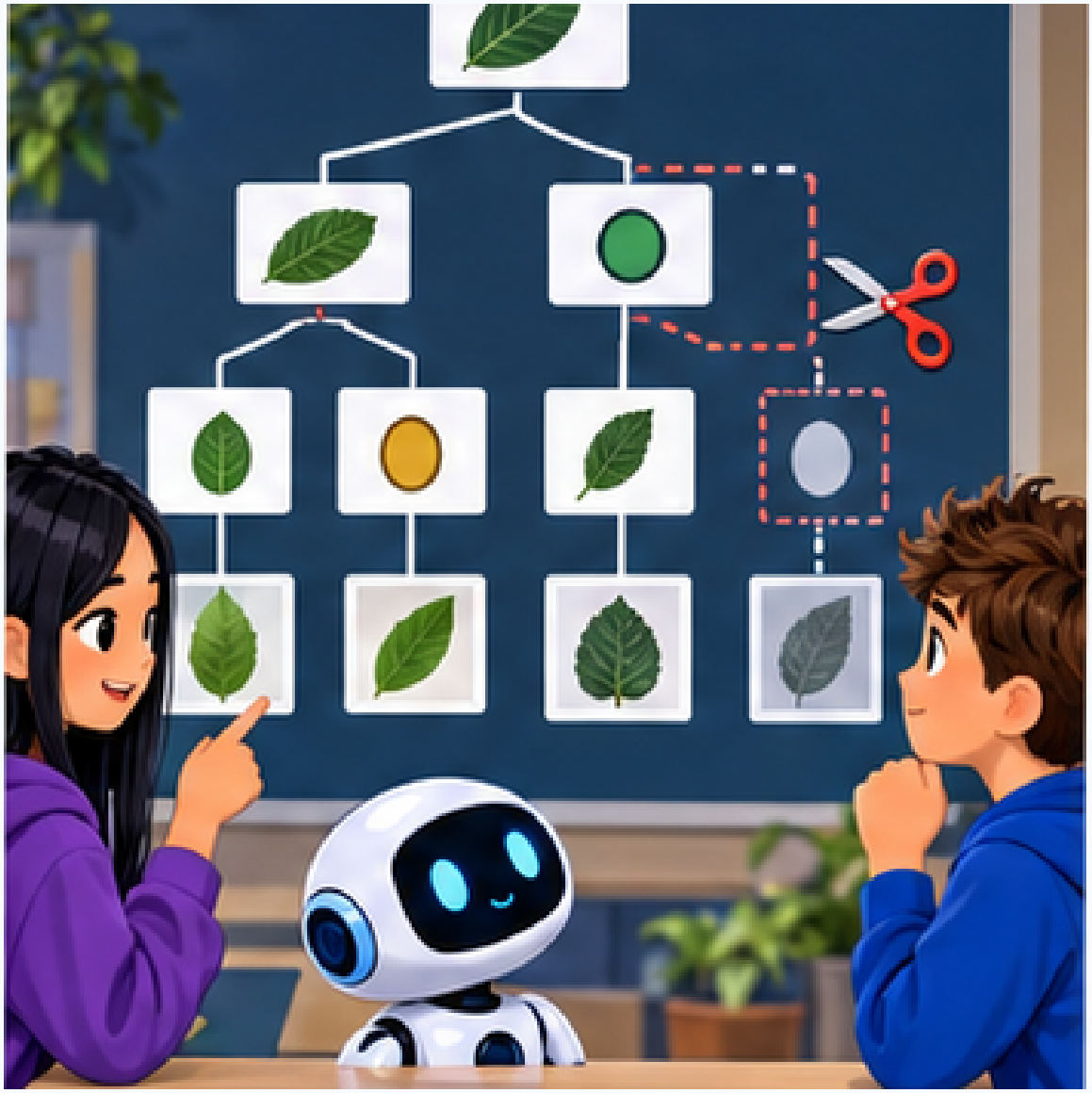
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.
</details>



LEARNING TARGET Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Read six rows before selecting a question

> Each candidate learns a majority label for each branch.

Depth-one scope and fictional source

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. Unlike the earlier fixed-threshold examples, where the two **output** labels were fixed in advance, this week we fit both the question and the label stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Complete near/tall training table

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear
5	0	0	clear
6	0	1	blocked

Candidate order, leaf tie and score tie policies

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group’s majority label; choose clear for tied label counts.

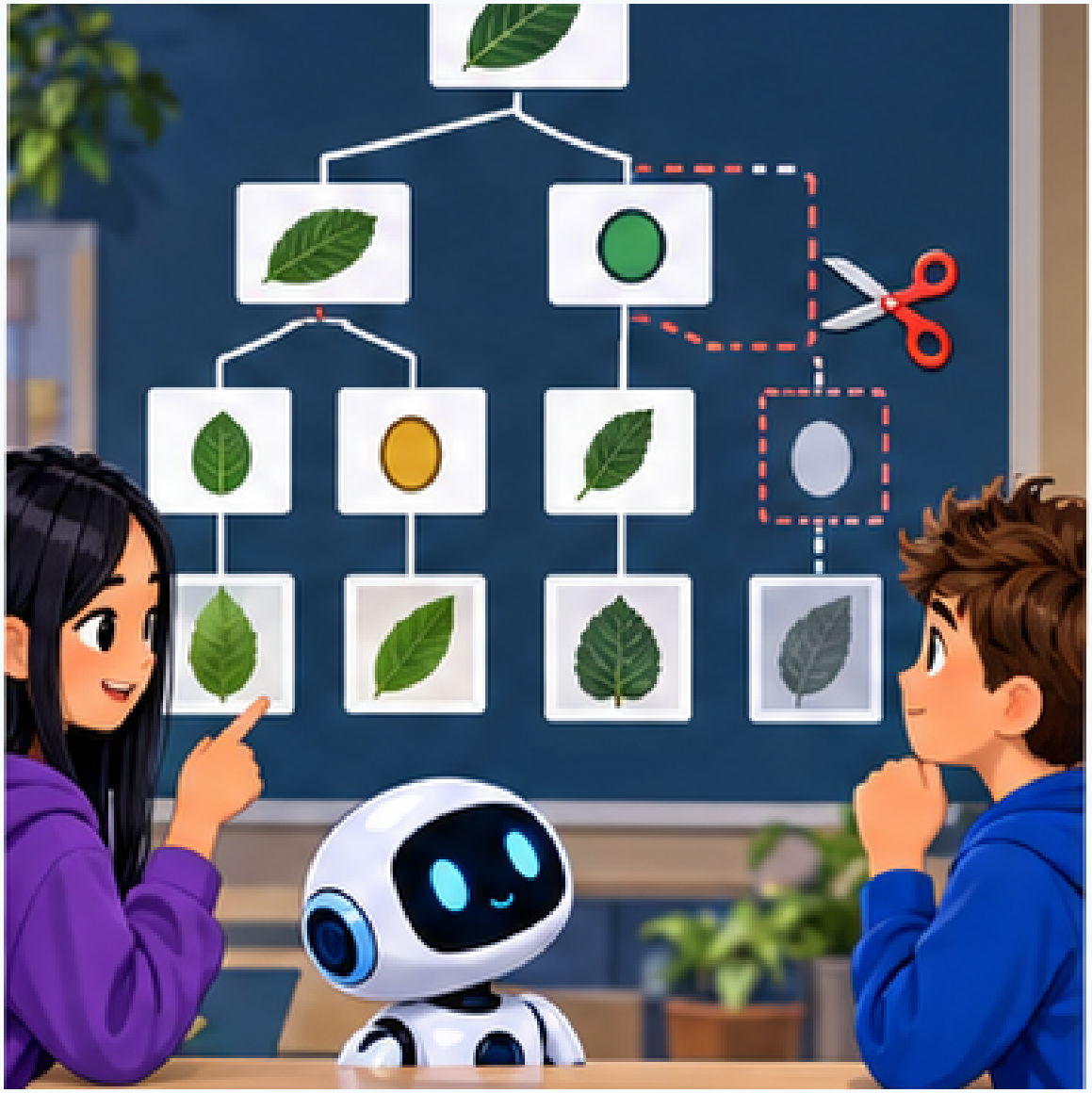
Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use decision trees to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Fit the feature and both leaf predictions

> Count mistakes across all rows rather than averaging branch percentages.

Complete depth-one fitting program

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Trace near1error versus tall2errors

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

State empty-branch assumptions and limited algorithm scope

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

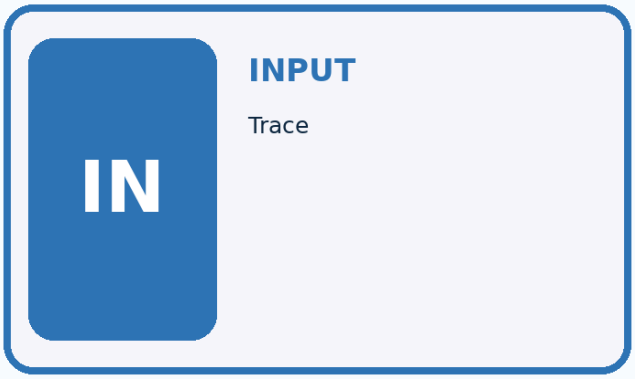
Keeping the scope small lets you inspect every decision.

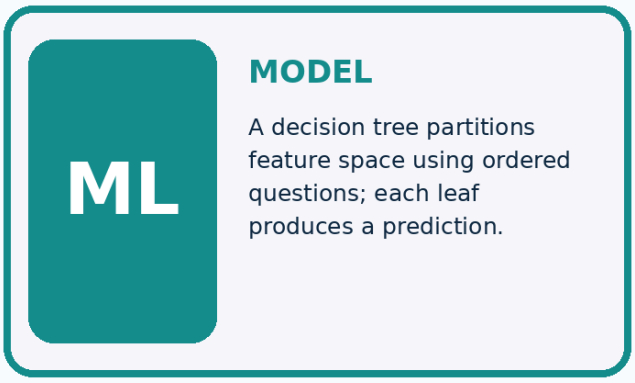
▼ **Earlier wording, alternatives and picture transcripts**

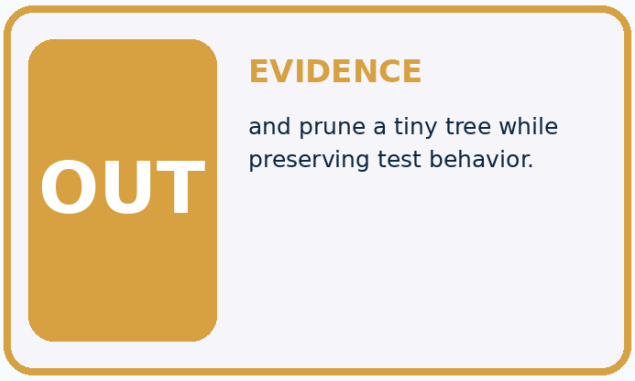
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Trace, draw, and prune a tiny tree while preserving test behavior.

MODEL IDEA A decision tree partitions feature space using ordered questions; each **leaf** produces a **prediction**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Root** asks whether spot_count ≥ 3. Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

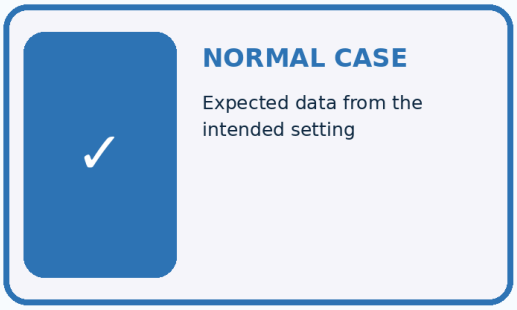
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

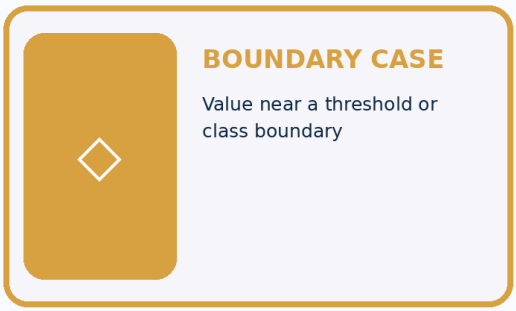
Compare a fitted decision with a supplied branch rule

> Boundary and missing-input checks explain inference; they do not establish training quality.

Four evaluation conditions



NORMAL Expected data and expected behavior.

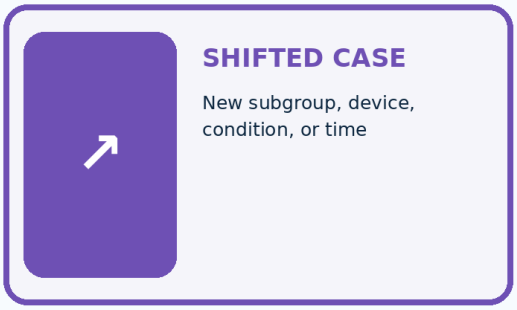


Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: The stored rule changes to visible_spots>2.

▼ **Check after predicting**

Prediction: M is now predicted healthy.

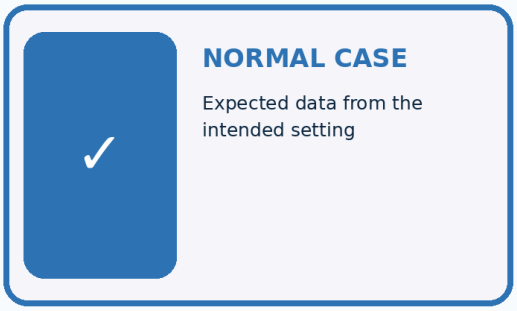
Reason: Changing >= to > moves the exact value 2 to the other **branch**.

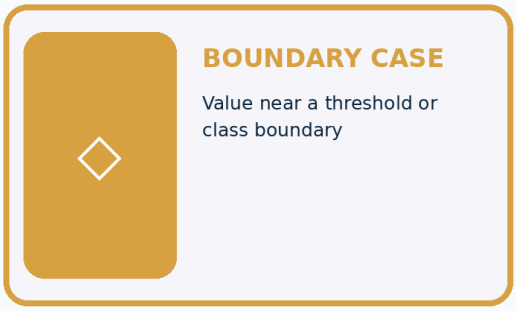
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

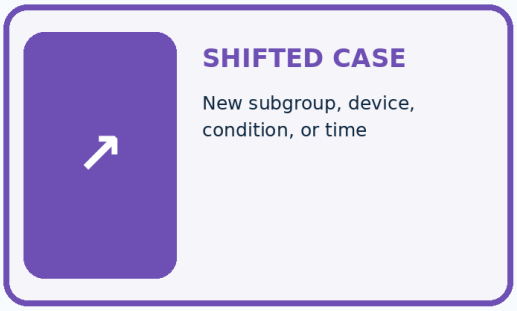
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace each branch and compare the actual missed rows

> Keep the fitted near/tall tree separate from the supplied spot-count picture.

Pictured spot inputs and scope

Trace a small fitted tree, including its exact boundary.

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

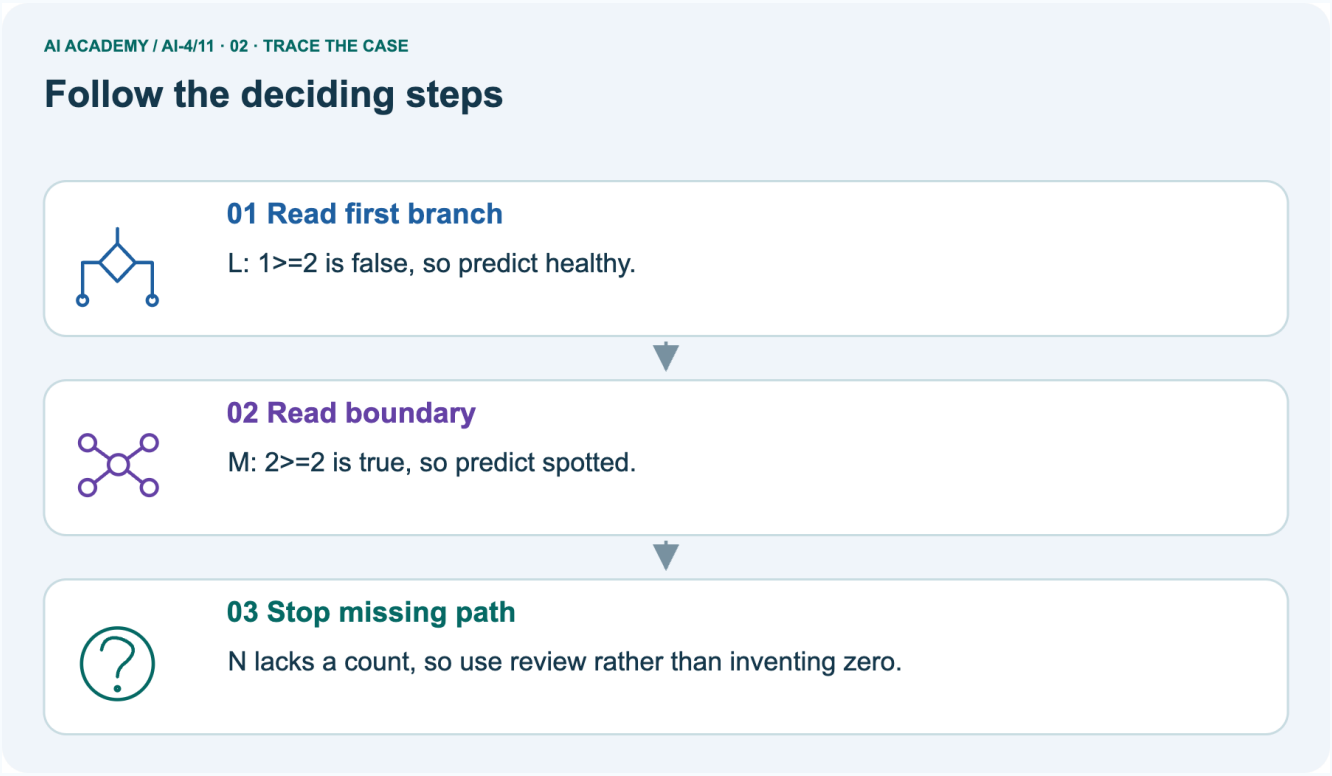
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main fitted tree chooses near and stores near0→clear,near1→blocked. On its four validation rows the tree missesrows3,4, while the blocked baseline missesrows2,4: both2/4but different errors. Separate picture uses a supplied visible_spots>=2 rule with review for missing; the

exact count2distinguishes>=2from>2. Neither picture nor boundary tracing proves that2was fitted or optimal.

The independent assessment uses its own cutoff5.

Trace ordinary, boundary and missing paths



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read first branch	L: 1>=2 is false, so predict healthy.
2. Read boundary	M: 2>=2 is true, so predict spotted.
3. Stop missing path	N lacks a count, so use review rather than inventing zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

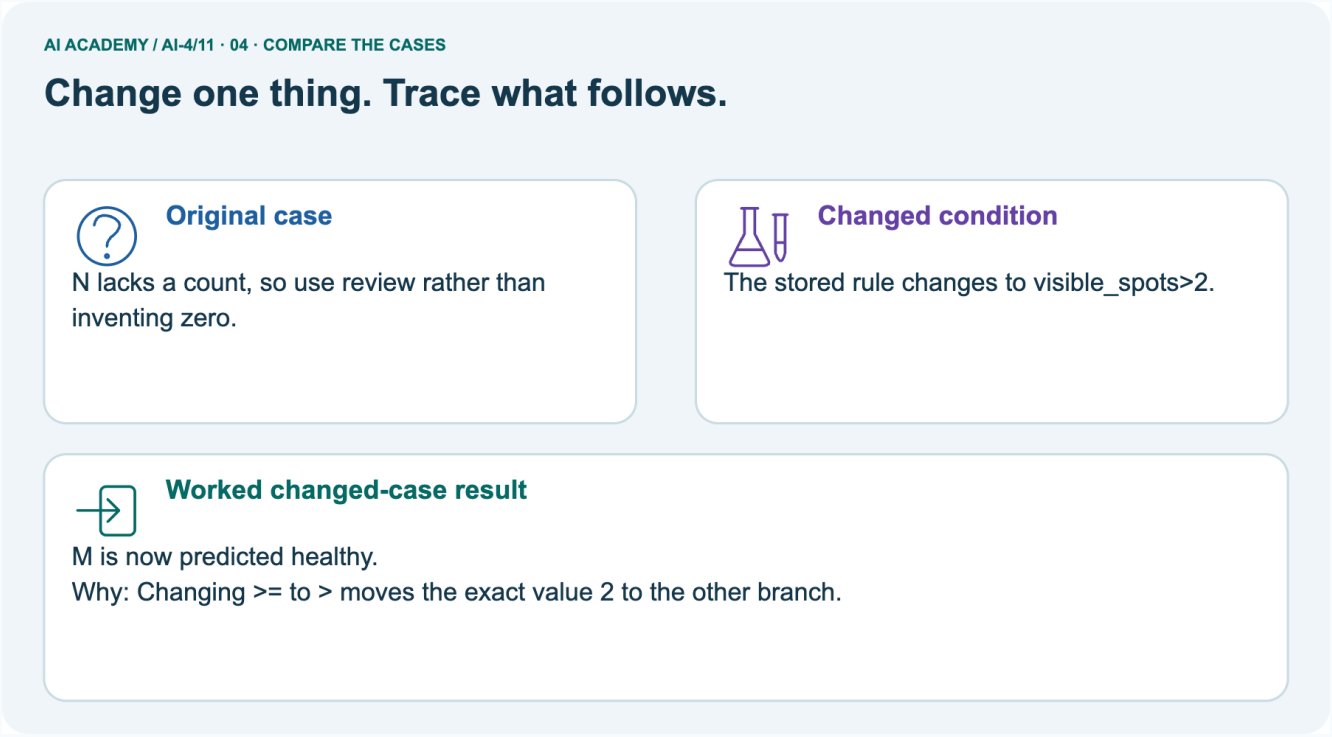
Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict>=2changingto>2

- Change one thing: The stored rule changes to visible_spots>2. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain inference versus fitting

- Applying a stored tree is inference. Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

- Read the labelled picture: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to learned near tree and validation comparison

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 11 • WORKED EVIDENCE

1. Root asks whether spot_count ≥ 3.
2. Yes branch predicts spotted; no branch asks whether yellow_ratio ≥ 0.25.
3. Trace leaf A: spot_count=1, yellow_ratio=0.10 → healthy.
4. A branch fitting one training oddity hurts test results, so prune and retest.

STEP 1 **Root** asks whether spot_count ≥ 3.

STEP 2 Yes **branch** predicts spotted; no **branch** asks whether yellow_ratio ≥ 0.25.

STEP 3 Trace **leaf** A: spot_count=1, yellow_ratio=0.10 → healthy.

STEP 4 A **branch fitting** one **training** oddity hurts test results, so prune and retest.

CHECK Verify the denominator, unit, **split**, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

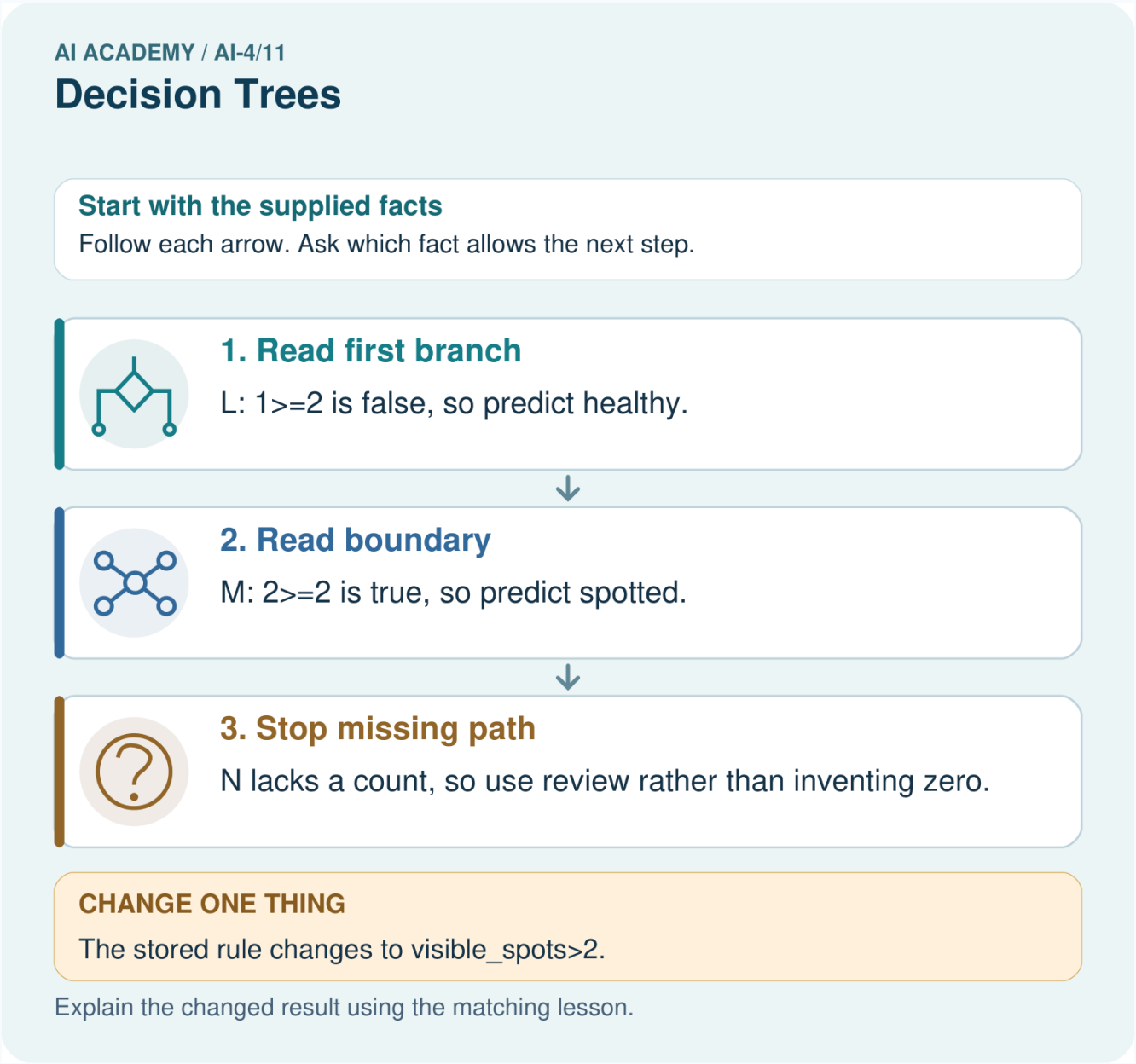
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a feature name mistaken for its learned answer

> Use the branch’s actual training labels.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A deeper tree must be better because it asks more detailed questions.

Keep error evidence and limits visible

- Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

  Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

- Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset.

  Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

> **A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A deeper tree must be better because it asks more detailed questions. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise candidate scores with less help

> Do not assume a near branch has an inevitable label.

Try W2 from the complete training table

Use workbook W2 for the connected example “**Fitting** a decision tree”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check learned labels and deeper-tree limits

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Challenge a fixed feature-to-answer assumption

- Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the exact pictured boundary

- Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the cutoff5independent task

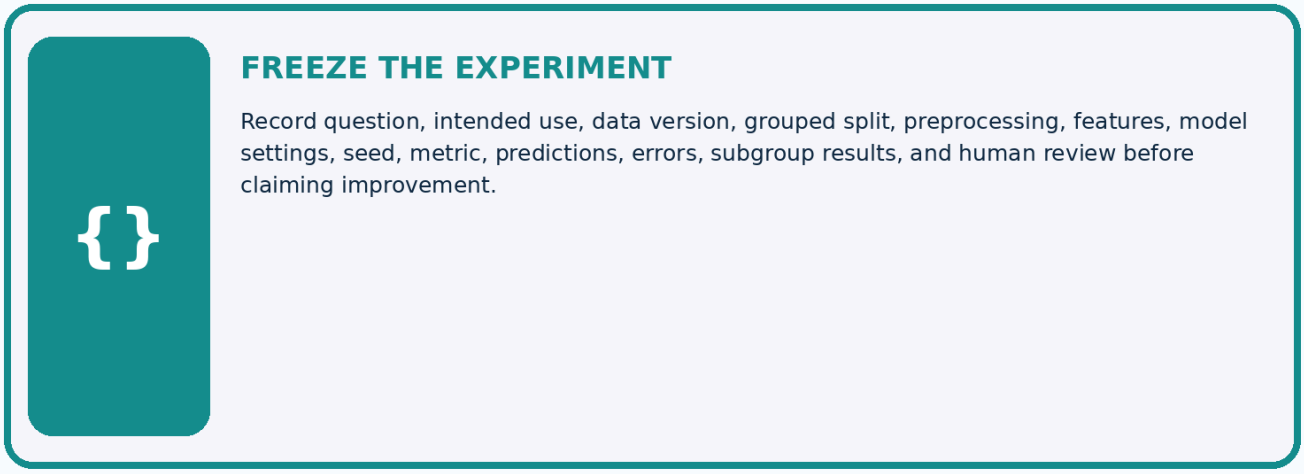
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success **threshold**

DATA version + grouped **split** + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Verify the candidate scores

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Build one complete depth-one tree and compare fixed predictions

> Every branch label and score is supported by the six training rows.

Complete branch-count trace

Candidate / branch	Training row labels	Learned leaf / errors
near=0	clear,clear,blocked	clear;1error
near=1	blocked,blocked,blocked	blocked;0errors
tall=0	blocked,blocked,clear	blocked;1error

Candidate / branch	Training row labels	Learned leaf / errors
tall=1	blocked,clear,blocked	blocked;1error
Chosen root	Near total1,tall total2	Near wins under declared total-mistake rule

Normal, binary-boundary, missing and refit probes

Complete core probe	Expected bounded check
Normal row near1,tall1	Fitted near tree predicts blocked.
Binary boundary near0,tall1	Stored near0leaf predicts clear; no continuous threshold is fitted here.
Missing near, known tall1	Original fit/inference examples require complete flags; request review, do not silently use tall or invent near0.
Changed training branch labels near0: [blocked,blocked,clear]	Refitting that branch under its majority policy gives blocked; a name alone does not fix the label.

Sort, fit, freeze and compare

1. Human Tree Builder: physically sort all six supplied training cards by each candidate flag. Learn each branch majority under clear-on-tie, count row mistakes and keep the first question if totals tie. Trace all six training predictions, then freeze the chosen near tree and compare its four supplied validation rows with the training-majority baseline. Record the different wrong-row identities despite equal accuracy.

   This main route fits a depth-one classroom learner, not a full library decision tree. The optional pruning comparison below uses separate supplied trees and development labels; it is not a final-test result. Keep the selected cutoff5task for an uncoached first attempt.

▼ **Optional supplied-tree pruning comparison**

Optional supplied-tree comparison on a separate development fixture	Complete facts
Full stored tree	If near0→clear; if near1then tall0→clear,tall1→blocked.
Pruned stored tree	If near0→clear; if near1→blocked.
Development rows in order	(near1,tall0,blocked),(1,1,blocked),(0,0,clear),(0,1,clear).
Trace before checking	Full predicts clear,blocked,clear,clear:3/4. Pruned predicts blocked,blocked,clear,clear:4/4.
Bound the conclusion	Pruning is preferred only on these supplied development rows; tree fitting history and final-test performance are not supplied. A later final check must stay separate.

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing

> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

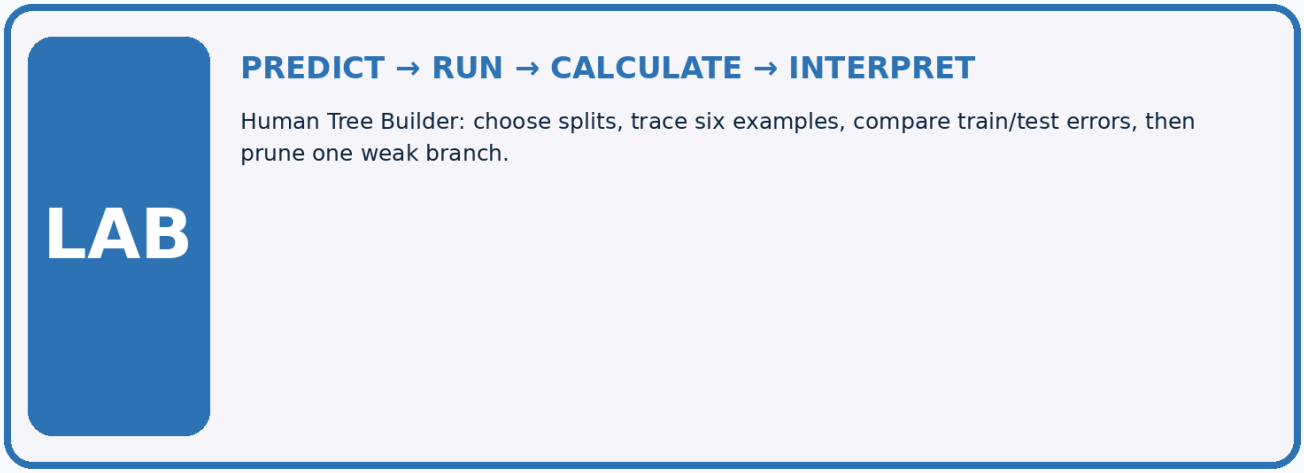
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Human Tree Builder: choose **split**s, trace six examples, compare train/test errors, then prune one weak **branch**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace the supplied cutoff5rule

> Apply the exact operator and keep missingness explicit.

Selected fresh task and scope

A stored fitted rule says spots >=5 predicts spotted; otherwise healthy. Trace4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

This task supplies a stored cutoff5and a missing-count review policy. It assesses branch tracing, equality and the fitting/evaluation distinction; it does not by itself assess fitting a new tree. State which evidence would be needed to claim training selected or improved the rule. Keep the earlier near/tall training and picture's cutoff2separate.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree predicts fern when **leaf**_width_cm <= 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a tree that another learner can trace and audit

> The project record must separate training choices from evaluation evidence.

Project artifact and learning record

THIS WEEK'S CHECKPOINT Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

Draw the fitted root, branch conditions, learned leaf labels and training counts. Keep the candidate order, leaf/score tie rules and stopping depth. Report row-level validation predictions and the same-set baseline with each denominator. Record missing-input review policy and any development pruning decision. Do not label a supplied rule as learned or reuse validation-driven changes as untouched final-test evidence.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, **split**, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

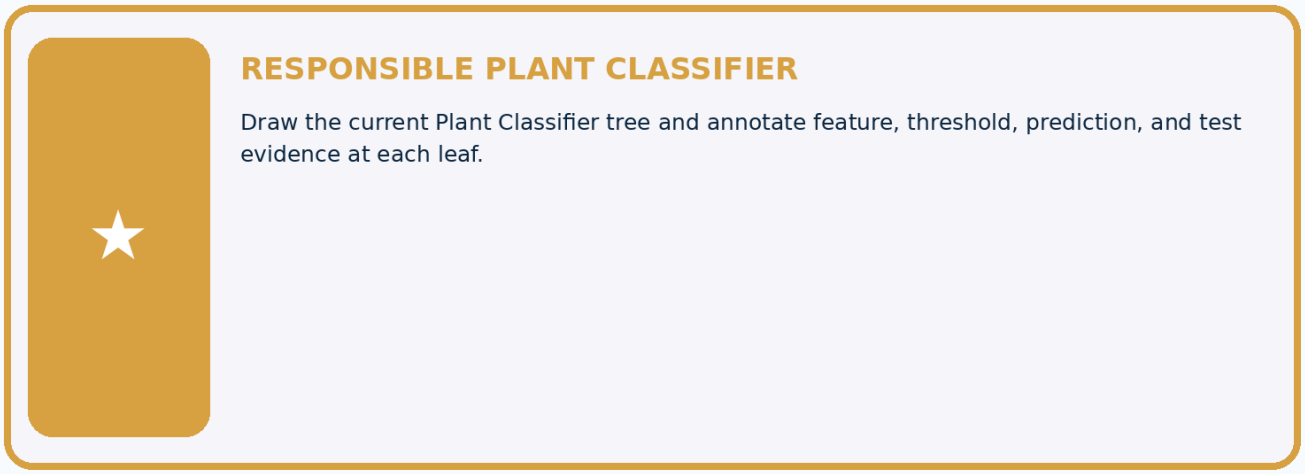
My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish by naming what was fitted and what was supplied

> Explain the evidence for one learned branch and one inference decision.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Fit a **root** question and **leaf predictions** from **training** examples

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
split	question dividing examples	
threshold	numeric boundary used in a split	
leaf node	terminal prediction	
pruning	remove unnecessary branches	

Closer-explanation terms

Word	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction

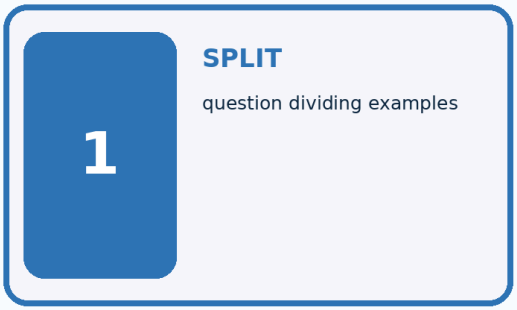
Word	Meaning
Depth	The largest number of question steps from root to leaf

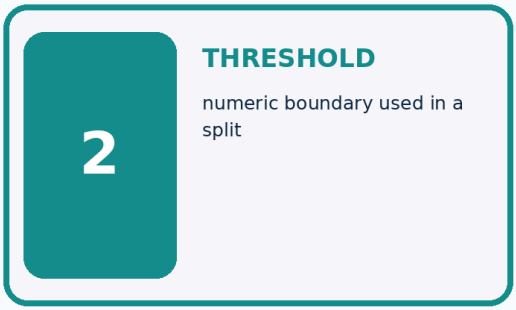
▼ Earlier wording, alternatives and picture transcripts

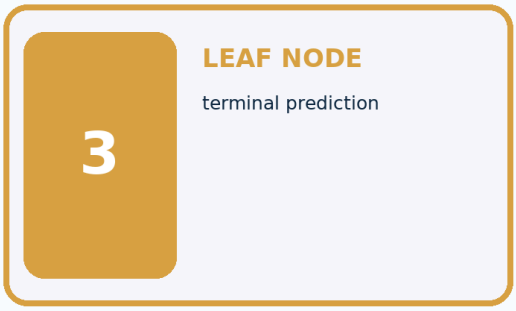
VOCABULARY

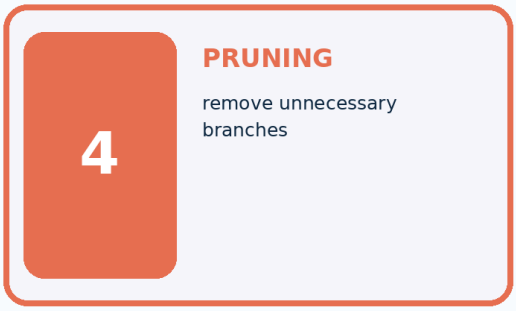
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A deeper tree must be better because it asks more detailed questions.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **split**, **threshold**, **leaf node**, **pruning**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A deeper tree must be better because it asks more detailed questions.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

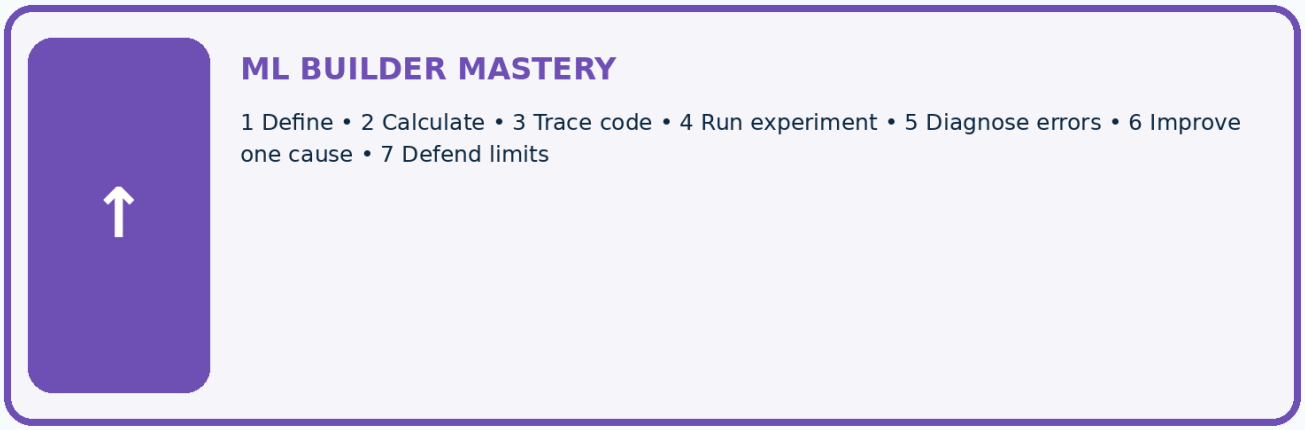
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the fresh stored-cutoff 5 boundary and missingness task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

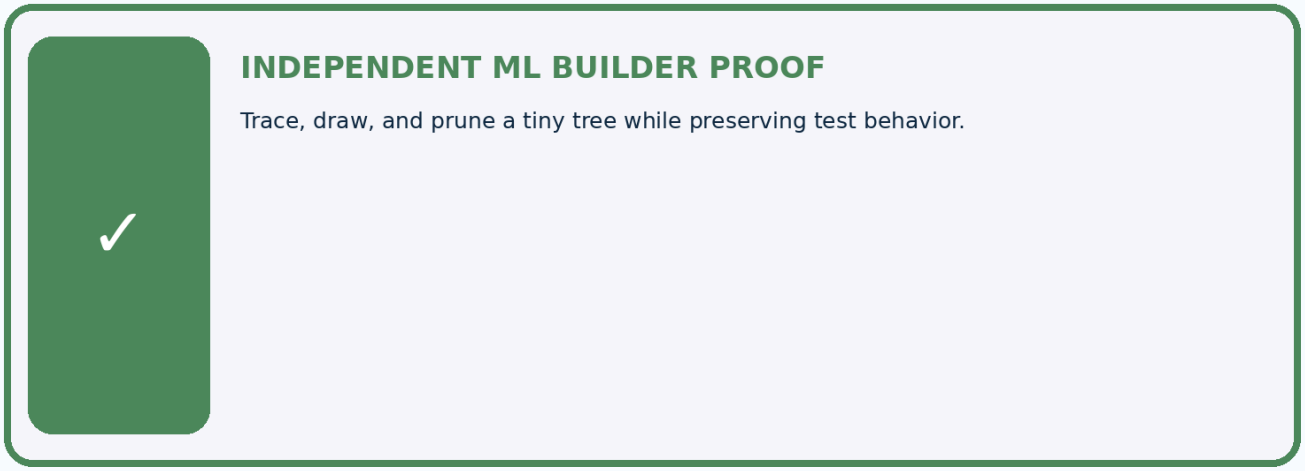
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Trace, draw, and prune a tiny tree while preserving test behavior.

NEXT BRIDGE Carry the experiment **evidence** into: Nearest Neighbors.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Trace 2,3,4 and state boundary.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

Evidence or stage	Value or result
2 spots	Below 3 → healthy
3 spots	At least 3 → spotted
4 spots	At least 3 → spotted

▼ Earlier wording, alternatives and picture transcripts

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Trace the inclusive tree threshold

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Trace 2,3,4 and state boundary.

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned **threshold**.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Fit a root question and leaf predictions from training examples	Explain the deciding step, show a trace or test, and state one limit.
Trace a new example through the learned tree	Explain the deciding step, show a trace or test, and state one limit.
Compare a shallow tree with a baseline and state its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Trace a small fitted tree, including its exact boundary.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Trace a small fitted tree, including its exact boundary.

- Which labelled step matches this detail? N lacks a count, so use review rather than inventing zero.
- 1. Read boundary
- 2. Read first **branch**
- 3. Stop missing path
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Stop missing path N lacks a count, so use review rather than inventing zero. Read the picture in order. N lacks a count, so use review rather than inventing zero.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The stored rule changes to visible_spots>2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.
- 3. M is now predicted healthy.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Repair and keep the record

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a **split** unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

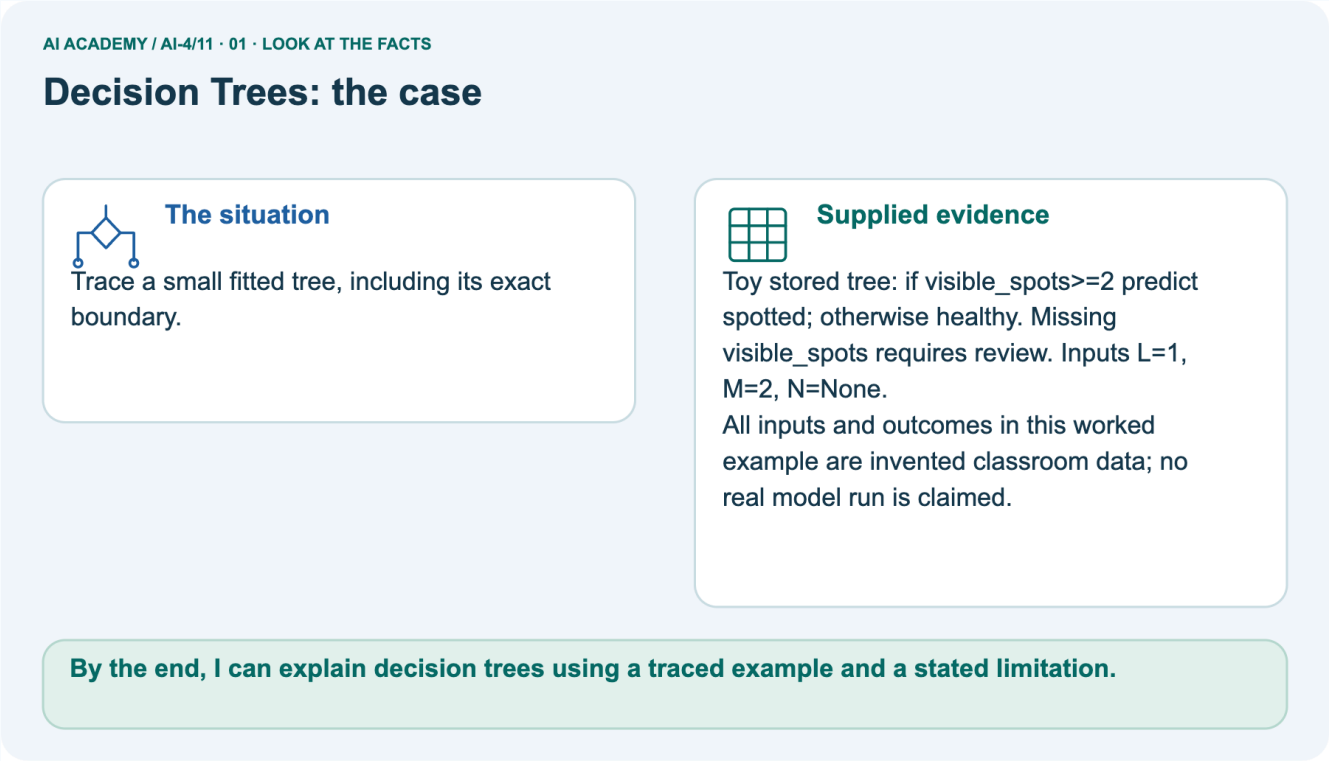
<details>
<summary>Reference · The six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Decision Trees: the case



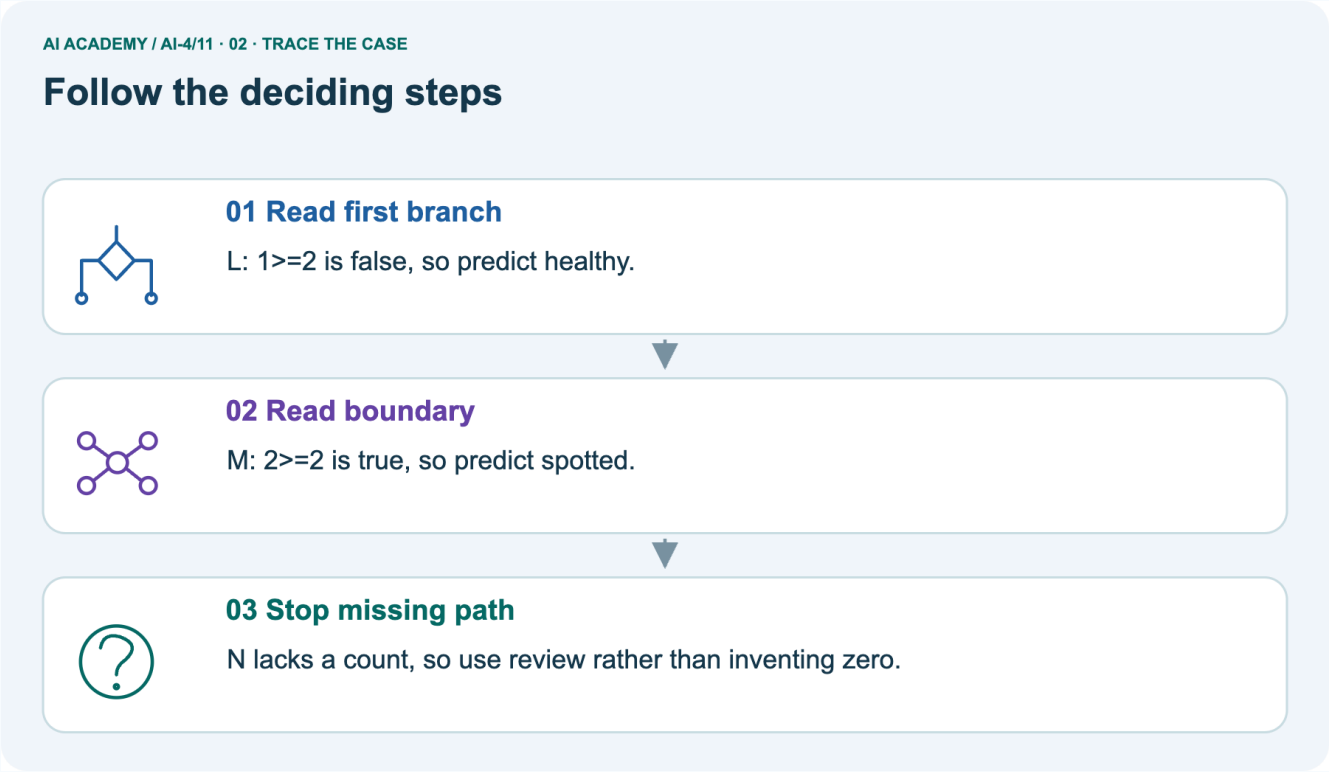
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace a small fitted tree, including its exact boundary. Supplied facts: Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

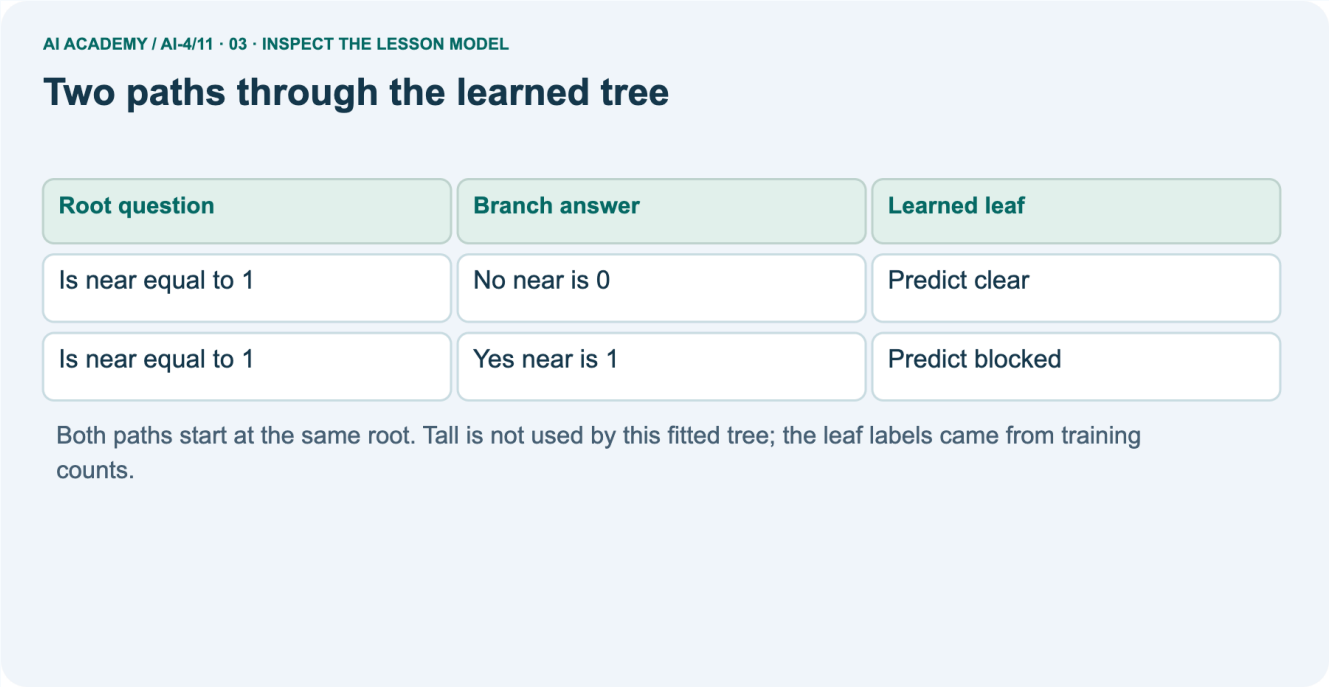


Follow the labelled picture with the matching explanation. Open the vector diagram

Read first **branch**: L: 1>=2 is false, so predict healthy. → Read boundary: M: 2>=2 is true, so predict spotted. → Stop missing path: N lacks a count, so use review rather than inventing zero.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two paths through the learned tree

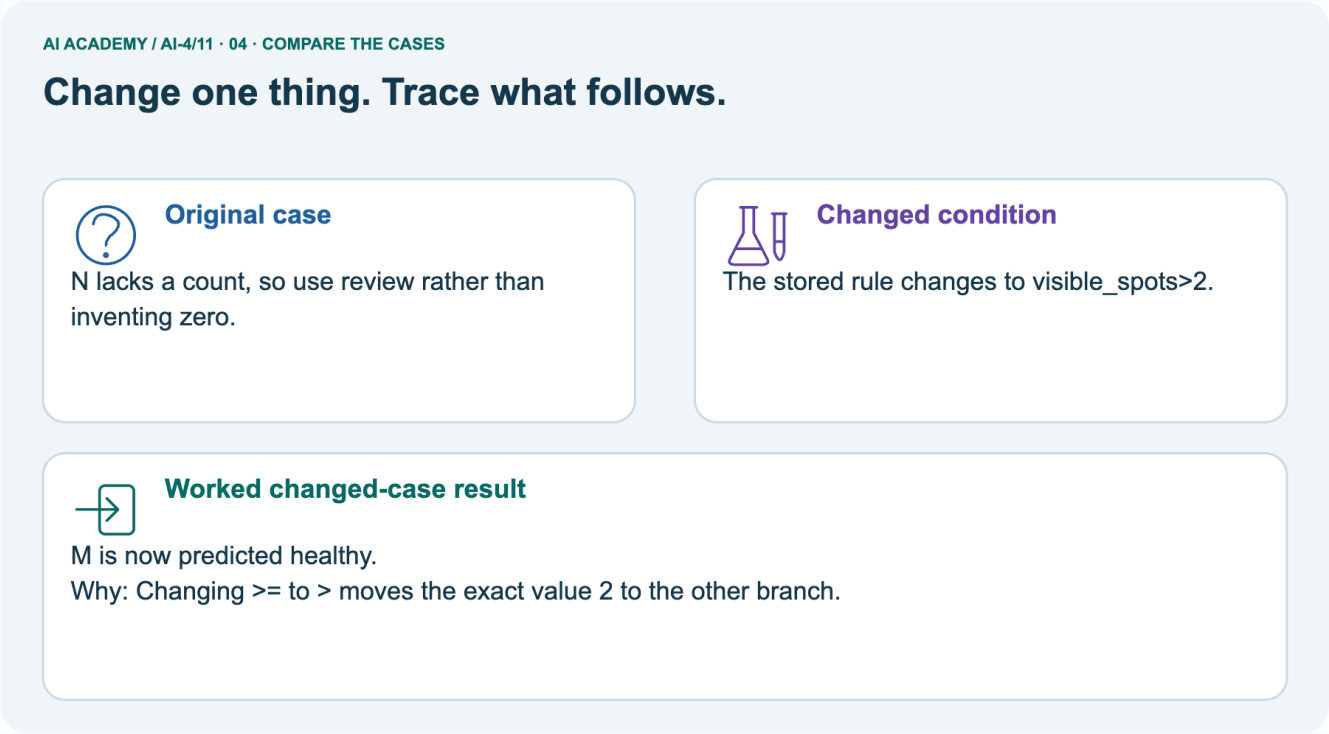


Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



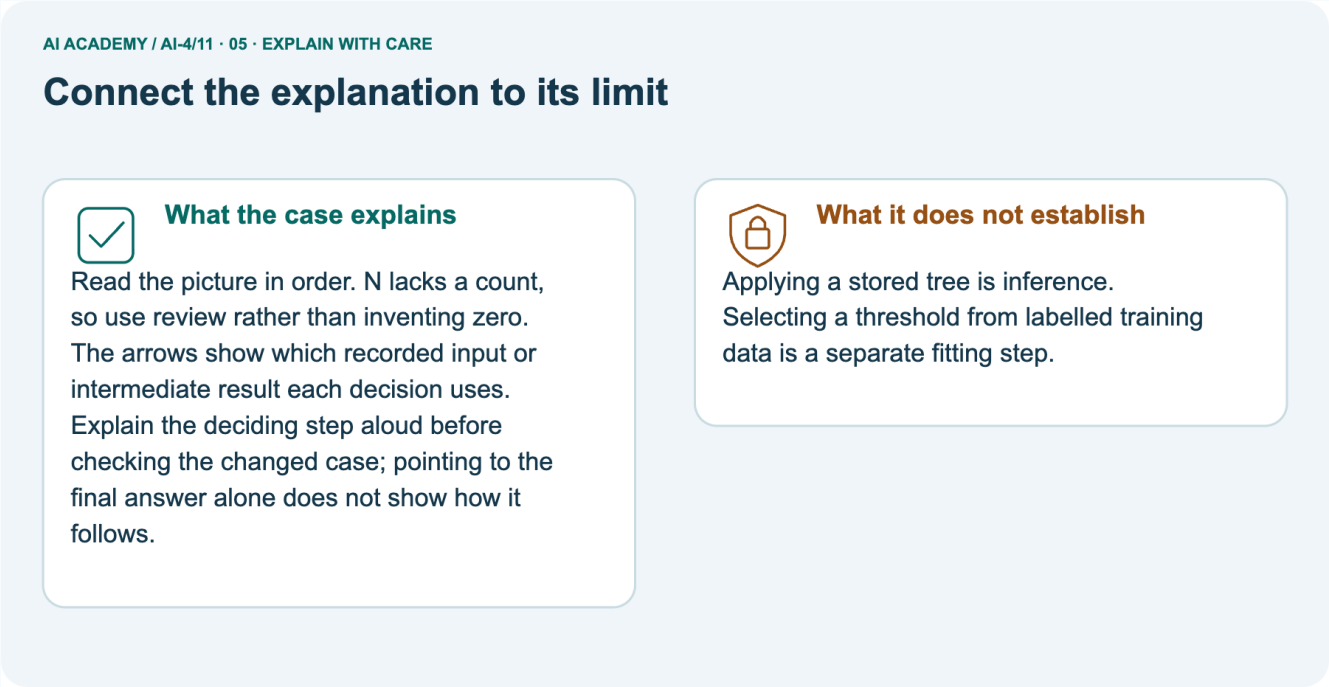
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N lacks a count, so use review rather than inventing zero. Changed condition: The stored rule changes to visible_spots>2. Expected result: M is now predicted healthy.

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

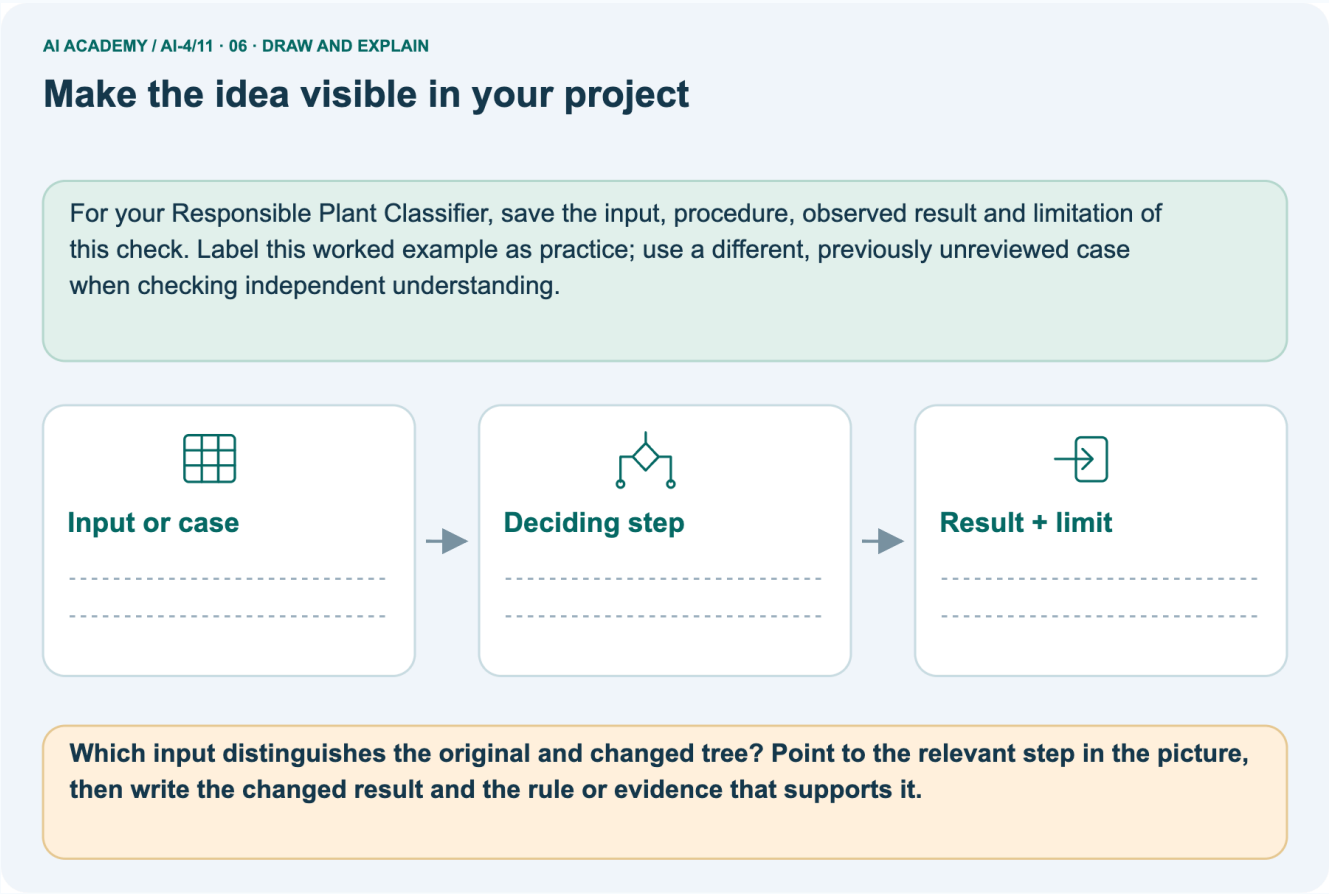
Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Applying a stored tree is inference.

Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.